

Kartik Raikar

AI Engineer — Generative AI — LLM Applications

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Atlas AI Resume](#)

Professional Summary

AI Engineer specializing in Generative AI, LLMs, RAG, and AI evaluation. Experienced in building production-oriented AI applications using Python, FastAPI, React.js, SQL, vector databases, and modern LLM tooling.

Projects

[Interactive AI Resume](#)

- AI Ops Root Cause Correlator – Incident Engine** | Python, FastAPI, PostgreSQL, Redis, NetworkX, Three.js
 - Engineered an autonomous incident correlation engine resolving cascading microservice alert storms in **0.78s** (down from 1–4 hours of manual tracing), achieving **100% Top-1 RCA accuracy** across 30 benchmark scenarios.
 - Implemented dynamic **EWMA anomaly detection** ($z > 2.0\sigma$) with causal DAG traversal in **NetworkX**, isolating multi-root-cause failures and achieving **100% precision/recall** in false-positive alert suppression.
 - Built a real-time 3D topology telemetry visualizer with **Three.js/WebGL** and streaming WebSockets, supporting counterfactual what-if blast radius simulation over **PostgreSQL 16 + pgvector**.
- VersionRAG – Documentation Intelligence** | Python 3.12, FastAPI, PostgreSQL, pgvector, React 18, TypeScript, Tailwind
 - Developed an enterprise RAG architecture resolving cross-version code contamination, eliminating hallucinated deprecated API calls from **62.5% to 0.0%** via database-enforced version-partitioned vector indexing.
 - Engineered a structure-aware **AST semantic diff engine** detecting undocumented breaking changes with **94.2% accuracy**, boosting retrieval precision @ k=6 from **41.7% to 98.4% (+136%)** at **5.4ms HNSW latency**.
 - Integrated a 4-step Chain-of-Version reasoning pipeline with live X-Ray diagnostic chunk inspection, Bcrypt-hashed OTP email auth, and resilient **pgvector** fallbacks.
- GitHub MCP Toolkit – Fault-Tolerant MCP Server** | Python, FastMCP, GitHub API, Ollama, Docker, GitHub Actions
 - Built a production-ready Anthropic **Model Context Protocol (MCP)** server for LLM-driven repository automation, semantic search, and deterministic issue triage with reversible write workflows.
 - Eliminated a measured **14% blind bulk-mutation rate** via a **two-phase SHA-256 preview-token protocol** requiring explicit human approval before mutating remote repository states.
 - Improved intent execution accuracy from **64% to 100%** and neutralized prompt injections across 20 adversarial test suites using **ABAC security**, circuit breakers, **Saga rollback**, and Pydantic validation.
- ApexRAG – RAG Retrieval Evaluation System** | Python, FastAPI, ChromaDB, Sentence Transformers, Ollama, scikit-learn
 - Built a comprehensive RAG evaluation benchmark over **2,580 documentation chunks** and 100 human-verified Q&A pairs on local CPU infrastructure to systematically isolate retrieval bottlenecks.
 - Benchmarked 5 retrieval strategies, improving retrieval accuracy from **61% (BM25)** to **85% (Cross-Encoder Re-ranking)**, establishing ranking quality as the primary bottleneck.
 - Eliminated data contamination and ID collisions using path-based hashing and deployed a **Logistic Regression query strategy router** for dynamic retrieval optimization.

Education

B.E. Computer Science & Engineering (AI & ML)

Jain College of Engineering, Belagavi

Aug 2023 – Jul 2027

CGPA: 8.50/10

Pre-University Course (PCM)

Jain PU College, Belagavi

2021 – 2023

80%

Technical Skills

Programming: Python, JavaScript, TypeScript, SQL

Databases: PostgreSQL, MySQL, MongoDB, Redis, Qdrant

Data & Visualization: Power BI, Tableau, Microsoft Excel

AI/ML: Generative AI, LLMs, RAG, Transformers, AI Evaluation, Prompt Engineering, Semantic Search, Vector Search

Frameworks: FastAPI, React.js, Next.js, PyTorch

Tools: Git, GitHub, Docker, Linux, Postman, VS Code

Certifications

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- AWS Training & Certification – Fundamentals of Machine Learning and Artificial Intelligence
- Tata – GenAI Powered Data Analytics Job Simulation (Forage)

Leadership & Achievements

- Vice President – Department of Artificial Intelligence & Machine Learning, Jain College of Engineering, Belagavi: Led departmental initiatives and coordinated technical and academic programs.
- Hack2Future 2.0 – IIIT Dharwad: Competed as part of **Team Velora** in a national-level hackathon.
- Code for Innovex – NITTE NMAM Institute of Technology: Participated in a **24-hour national-level hackathon**.

Relevant Coursework

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Data Science, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Machine Learning